

Project Planning

Phase: Project Planning

Project: ShopEEZ | Date: June 2026 | Stack: MongoDB · Express · React · Node.js

1. Project Goal

Deliver a fully functional, production-ready MERN stack e-commerce web application with customer-facing shopping features and a complete admin management dashboard within a 5-week development timeline.

2. Development Milestones

Phase	Milestone	Target	Status
Phase 1 — Setup	Vite React + Express API + MongoDB Atlas connection	Week 1	Complete
Phase 2 — Auth	User register/login, JWT middleware, bcrypt hashing	Week 1	Complete
Phase 3 — Products	Product schema, CRUD APIs, catalog with search & filters	Week 2	Complete
Phase 4 — Cart	Cart schema, Redux cartSlice, LocalStorage persistence	Week 2	Complete
Phase 5 — Orders	Checkout flow, Order schema, placement, order history	Week 3	Complete
Phase 6 — Admin	Admin dashboard, product/order/user management, stats API	Week 3	Complete
Phase 7 — UI Polish	Dark theme CSS, animations, responsive design	Week 4	Complete
Phase 8 — Wishlist	Wishlist Redux slice, wishlist page, heart icon on cards	Week 4	Complete
Phase 9 — Reviews	Product review form, star rating, average rating	Week 4	Complete
Phase 10 — Docs	MERN Phase Wise docs, README, project plan	Week 5	Complete

3. Sprint Breakdown

Sprint 1 — Foundation (Week 1)

- Setup monorepo structure (client/ + server/)
- Configure Vite + React 19 + React Router v6
- Configure Express server with ES Modules
- Connect MongoDB Atlas via Mongoose
- Create User Mongoose model with bcrypt pre-save hook
- Implement POST /api/auth/register and POST /api/auth/login
- Create JWT protect middleware
- Create Redux authSlice with localStorage token persistence
- Build Login and Register pages

Sprint 2 — Core Shopping (Week 2)

- Create Product Mongoose model
- Implement GET /api/products with search, category, price, sort params
- Create Redux productSlice for catalog state
- Build Home.jsx, ProductList.jsx, ProductDetail.jsx
- Create Redux cartSlice with LocalStorage sync
- Build Cart.jsx with quantity controls and total calculation (Rs + GST)
- Build ProductCard.jsx with hover animations

Sprint 3 — Checkout & Admin (Week 3)

- Create Order Mongoose model
- Implement POST /api/orders — create order, decrement stock
- Create admin middleware (role check)
- Build Admin Dashboard with stats cards
- Build Admin Products management (form + table + CRUD)
- Build Admin Orders management (status toggle)
- Build Admin Users management (list + delete)

4. Risk Register

Risk	Probability	Impact	Mitigation
MongoDB Atlas connectivity	Low	High	Use local MongoDB as fallback during development
JWT token expiry breaking UX	Low	Medium	Axios interceptor — 401 response redirects to /login
Image uploads failing on deploy	Medium	Medium	Local /uploads/ dir with Cloudinary as production fallback
Redux state bloat	Low	Low	Paginate product API responses (10 per page)